Iqbal Habib
Masters in Applied Statistics and Data Science, Jahangirnagar University, Bangladesh.
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ResearchGate | GitHub | LinkedIn

RESEARCH INTEREST

AI in Healthcare, Computer Vision, Image processing, Data Science, Machine Learning (ML), Deep Learning (DL), Data analysis, Statistical modelling, Mathematical modeling, Bioinformatics, Biomedical engineering

ACADEMIC BACKGROUND

Masters in Applied Statistics and Data Science *2022-2023*
Jahangirnagar University
Savar, Dhaka-1342, Bangladesh.
CGPA: **3.92** on a scale of **4.00 (3.97/4.00, WES)**
Data Analysis project: *Unveiling Crime Dynamics in Bangladesh: A machine learning odyssey from clustering to prediction.*
Supervisor: **Md. Rezaul Karim**, PhD (KU Leuven & UHasselt), MS (Biostatistics), MS (Statistics)

- Applied PCA, clustering (K-means, Hierarchical), and predictive models (Random Forest, Decision tree ANN, SVM) to analyze crime patterns in Bangladesh Crime Data (2010–2019).
- Random Forest model achieved best performance ($R^2 = 0.995$), supporting evidence-based law enforcement resource allocation through clustering and forecasting techniques.

Bachelor of Science in Mathematics *2013- 2017*
National University
Gazipur -1704, Bangladesh
CGPA: 2.97 on a scale of 4.00 (average 3.08 in the last 100 credit) (2.85/4.00, WES)

RESEARCH EXPERIENCE

Senior Research Associate *December 2024 – Present*
Data Insightopia, Bangladesh

- Analyzed data for research in data science, focusing on machine learning, AI, and data modeling.
- Worked with diverse teams to design and implement innovative data analysis projects.
- Built predictive models and recommendation systems to enhance strategic decision-making and user engagement.

Research Assistant *November 2023 – Present*
Department of Computer Science and Engineering,
Pabna University of Science and Technology, Bangladesh
Supervisor: Nakib Aman

- Assisted in research on medical image classification and AI in healthcare.
- Performed data preprocessing, augmentation, and analysis for medical imaging datasets.
- Contributed to three IEEE conference papers on cancer prediction using hybrid deep learning models.
- Prepared figures, results, and manuscript sections for publication.
- Supported report writing, presentation preparation, and project documentation.

Research Intern *April 2023 – August 2023*
Advanced Machine Intelligence Research Lab-AMIRL, Bangladesh

- Prepared data collected in KOBO for analysis by performing complex data preprocessing tasks
- Assisted with preparing project-related reports, presentations and manuscripts.

Associate Data Analyst *March 2020 – March 2023*
Islam Jahid & Co.

- Led the implementation and optimization of an MIS system, automating financial reporting with Power BI, improving efficiency by 30%.
- Developed predictive models using time-series forecasting (ARIMA, LSTM) to enhance stock price predictions and provide actionable insights.

TEACHING EXPERIENCE

Private Mathematics Tutor (Grades 10–12) *2014 – Present*
Dhaka, Bangladesh

- Taught over 150 students individually and in small groups from the National Curriculum of Bangladesh, Edexcel, and Cambridge IGCSE/A Levels and Australian Curriculum.
- Covered core and advanced topics in Algebra, Calculus, Trigonometry, Statistics, and Geometry.
- Maintained consistent track record of student success, with significant improvement in grades and exam performance.
- Balanced tutoring with full-time research work, continuing to mentor select students after regular working hours.

TESTS

International English Language Testing System (IELTS) *19th October 2024*
Overall Score 6.5, Listening 7.5, Reading 6.5, Writing 6, Speaking 6
Graduate Record Exam (GRE) *30th January 2026*
Quant - 162, Verbal - 149

PUBLICATIONS

Peer-reviewed Conference Papers (IEEE)

[1] M. M. H. Munna, **I. Habib** and N. Aman, **"A Novel Hybrid Xception-DenseNet121 Model for Acute Lymphoblastic Leukemia Classification,"** *2024 6th International Conference on Sustainable Technologies for Industry 5.0 (STI)*, Narayanganj, Bangladesh, **2024**, pp. 1-6,doi: 10.1109/STI64222.2024.10951150

[2] **I. Habib**, M. M. H. Munna and N. Aman**, "Leveraging Hybrid Deep Learning for Effective Multiclass Blood Cancer Detection,"** *2024 International Conference on Recent Progresses in Science, Engineering and Technology (ICRPSET)*, Rajshahi, Bangladesh, **2024**, pp. 1-5,doi: 10.1109/ICRPSET64863.2024.10955897

[3] M. M. H. Munna**, I. Habib**, N. A. Turzo, N. Jahan and S. Nusrat, **"Automated Diagnosis of Acute Lymphoblastic Leukemia Leveraging EfficientNet and Fastai,"** *2024 27th International Conference on Computer and Information Technology (ICCIT)*, Cox's Bazar, Bangladesh, **2024,** pp. 233-238, doi: 10.1109/ICCIT64611.2024.11021950

Journal Publications

[4] Biva Anika Tahsin, ABM Shahadat Hossain, Md Shafiul Alom Khan, and **Iqbal Habib. "Benchmarking Hybrid ANN-LSTM and Physics-Informed Neural Networks for Forecasting Stock Market Prices."** *International Journal of Computer Applications.***2024***,* 975: 8887, doi: 10.5120/ijca2025924624

[5] Biva, Anika Tahsin, ABM Shahadat Hossain, Md Shafiul Alom Khan, and **Iqbal Habib. "Enhancing Financial Time Series Predictions with a Hybrid BNN-LSTM Approach,"** *International Journal of Applied Information Systems,* **2024***,* 12(47):1-7, doi: 10.5120/ijais2025452015

Under Review

[6] Md. Kamrul Hasan, Akher Ali, S M Junaid, S M Junaid**, Iqbal Habib**, Mohammad Alamgir Kabir.**"Multilevel Analysis of Socio-Demographic Determinants of Women's Empowerment in Household Decision-Making in Bangladesh: Evidence from BDHS 2022"** *BMC Women's Health* **2025**

[7] Mohammad Mehedi Hasan Munna, Nakib Aman**, Iqbal Habib**, Sabiha Nusrat, and Smita Saha. **"Evaluating the Impact of Lossy and Lossless Compression on Medical Image Classification Using Hybrid Deep Learning Models with Explainable AI,"** *Technologies* **2026**

TECHNICAL SKILLS

- **Programming language:** *Python, R, HTML, CSS*
- **Libraries & Frameworks:** *Pandas, Numpy, Scipy, Scikit-learn, Matplotlib, Seaborn, plotly, TensorFlow, Keras, Sklearn.metrics, Statsmodels*
- **ML & Data Science:** *Data Cleaning, Feature Engineering, Sampling, Visualization, Ensemble Methods, Model Evaluation, Cross-Validation, Image Classification*
- **Algorithms:** *Linear Regression, Logistic Regression, Decision Tree, Random Forest, SVM, ANN, CNN, BNN, PINN, LSTM, XGBoost*
- **CNN Architectures:** *Xception, ResNet50, DenseNet121, EffucientNet, VGG, GoogleNet*
- **Statistical & Analytics:** *SPSS, Excel, Google Sheets*
- **Development Tools:** *Jupyter Notebook, Google Colab, Kaggle, Github, LateX, Draw.io*

ACADEMIC RESEARCH AND PROJECTS

[1] Parkinson's Disease Detection Using Machine Learning and Deep Learning

- Developed a predictive model using XGBoost and Neural Networks to detect Parkinson's disease based on voice-related features.
- Achieved 94.12% accuracy and 0.96 F1 score, demonstrating effective early detection potential.

[2] Development of a CNN Model for Tuberculosis Detection Using Chest X-rays

- Developed a convolutional neural network (CNN) model using multiple convolutional and max-pooling layers for feature extraction, with dropout regularization to prevent overfitting, Adam optimizer and SparseCategoricalCrossentropy loss for multi-class classification to detect tuberculosis from chest X-rays, achieving an accuracy of 98.43%.
- Applied data augmentation techniques (SMOTE) to balance the dataset and enhance model performance.
- Achieved high precision (99.55%), recall (97.24%), and F1 score (98.38%), demonstrating effective TB detection capability.

[3] Titanic Survival Prediction using Machine Learning

- *Data Cleaning:* Handled missing values, outliers, and feature engineering.
- *Modeling:* Applied Logistic Regression, Decision Trees, Random Forest, and XGBoost; Random Forest achieved 85.56% accuracy.
- *Automation:* Built an automated pipeline for data preprocessing, encoding, and model training

Also explored different ML and DL algorithm for various projects as course work and personal interest.

ACADEMIC ACHIEVEMENTS AND AWARDS

- Won the best paper award at the *ICCIT* 2024, Cox's Bazar, Bangladesh,conference for presenting , **"Automated Diagnosis of Acute Lymphoblastic Leukemia Leveraging EfficientNet and Fastai,"** paper.
- Best poster presentation in graduate (Masters) Project presentation at Jahangirnagar University
- Awarded special mention for reaching into the final at Mathematical quize 2023 campus round.
- Jahangirnagar University academic performance award -2024
- Ministry of Education Bangladesh Jashore Board Scholarship for Secondary School Certificate Examination (52nd merit)- 2010

CERTIFICATION

- Data analysis using SPSS and R by Jahangirnagar University
- Research Learning Program 2024 by Bangladesh Research Society (BDRS)
- Data Science with Python Level 1 & 2
- AI for Medical Diagnosis- an online non-credit course authorized by Deep Learning AI.
- Mathematics for Machine Learning and Data Science
- Computer Architecture by Princeton University
- Deep Learning Specialization by Coursera

PROFESSIONAL MEMBERSHIPS

- Institute of Electrical and Electronics Engineers (IEEE) *2023 - Present*

REFERENCES

Md. Rezaul Karim, Ph.D (KU Leuven & UHasselt)
(M.Sc. Supervisor)
Professor
Department of Statistics and Data Science
Jahangirnagar university
Savar, Dhaka-1342, Bangladesh.
Email: rezaul@juniv.edu

Nakib Aman (Research supervisor)
Lecturer
Department of Computer Science and Engineering
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Thank you